Lecture 1

Systems of Linear Equations

In this lecture, we will introduce linear systems and the method of row reduction to solve them. We will introduce matrices as a convenient structure to represent and solve linear systems. Lastly, we will discuss geometric interpretations of the solution set of a linear system in 2- and 3-dimensions.

1.1 What is a system of linear equations?

Definition 1.1: A **system of m linear equations** in n unknown variables $x_1, x_2, \ldots, x_n$ is a collection of m equations of the form

$$\begin{array}{ccccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & a_{13}x_3 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & a_{23}x_3 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ a_{31}x_1 & + & a_{32}x_2 & + & a_{33}x_3 & + & \cdots & + & a_{3n}x_n & = & b_3 \\ \vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & a_{m3}x_3 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array} \tag{1.1}$$

The numbers a_{ij} are called the **coefficients** of the linear system; because there are m equations and n unknown variables there are thefore $m \times n$ coefficients. The main problem with a linear system is of course to solve it:

Problem: Find a list of n numbers $(s_1, s_2, \ldots, s_n)$ that satisfy the system of linear equations (1.1).

In other words, if we substitute the list of numbers $(s_1, s_2, \ldots, s_n)$ for the unknown variables $(x_1, x_2, \ldots, x_n)$ in equation (1.1) then the left-hand side of the ith equation will equal b_i. We call such a list $(s_1, s_2, \ldots, s_n)$ **a solution** to the system of equations. Notice that we say "a solution" because there may be more than one. The **set** of all solutions to a linear system is called its **solution set**. As an example of a linear system, below is a linear

system consisting of $m = 2$ equations and $n = 3$ unknowns:

$$\begin{aligned} x_1 - 5x_2 - 7x_3 &= 0 \\ 5x_2 + 11x_3 &= 1 \end{aligned}$$

Here is a linear system consisting of $m = 3$ equations and $n = 2$ unknowns:

$$\begin{aligned} -5x_1 + x_2 &= -1 \\ \pi x_1 - 5x_2 &= 0 \\ 63x_1 - \sqrt{2}x_2 &= -7 \end{aligned}$$

And finally, below is a linear system consisting of $m = 4$ equations and $n = 6$ unknowns:

$$\begin{aligned} -5x_1 + x_3 - 44x_4 - 55x_6 &= -1 \\ \pi x_1 - 5x_2 - x_3 + 4x_4 - 5x_5 + \sqrt{5}x_6 &= 0 \\ 63x_1 - \sqrt{2}x_2 - \tfrac{1}{5}x_3 + \ln(3)x_4 + 4x_5 - \tfrac{1}{33}x_6 &= 0 \\ 63x_1 - \sqrt{2}x_2 - \frac{1}{5}x_3 - \tfrac{1}{8}x_4 - 5x_6 &= 5 \end{aligned}$$

Example 1.2. Verify that $(1, 2, -4)$ is a solution to the system of equations

$$\begin{aligned} 2x_1 + 2x_2 + x_3 &= 2 \\ x_1 + 3x_2 - x_3 &= 11. \end{aligned}$$

Is $(1, -1, 2)$ a solution to the system?

Solution. The number of equations is $m = 2$ and the number of unknowns is $n = 3$. There are $m \times n = 6$ coefficients: $a_{11} = 2$, $a_{12} = 1$, $a_{13} = 1$, $a_{21} = 1$, $a_{22} = 3$, and $a_{23} = -1$. And $b_1 = 0$ and $b_2 = 11$. The list of numbers $(1, 2, -4)$ is a solution because

$$2 \cdot (1) + 2(2) + (-4) = 2$$

$$(1) + 3 \cdot (2) - (-4) = 11$$

On the other hand, for $(1, -1, 2)$ we have that

$$2(1) + 2(-1) + (2) = 2$$

but

$$1 + 3(-1) - 2 = -4 \neq 11.$$

Thus, $(1, -1, 2)$ is not a solution to the system. □

A linear system may not have a solution at all. If this is the case, we say that the linear system is **inconsistent**:

INCONSISTENT ⇔ NO SOLUTION

A linear system is called **consistent** if it has at least one solution:

CONSISTENT ⇔ AT LEAST ONE SOLUTION

We will see shortly that a consistent linear system will have either just one solution or infinitely many solutions. For example, a linear system cannot have just 4 or 5 solutions. If it has multiple solutions, then it will have infinitely many solutions.

Example 1.3. Show that the linear system does not have a solution.

$$\begin{aligned} -x_1 + x_2 &= 3 \\ x_1 - x_2 &= 1. \end{aligned}$$

Solution. If we add the two equations we get

$$0 = 4$$

which is a contradiction. Therefore, there does not exist a list (s_1, s_2) that satisfies the system because this would lead to the contradiction $0 = 4$. □

Example 1.4. Let t be an arbitrary real number and let

$$\begin{aligned} s_1 &= -\tfrac{3}{2} - 2t \\ s_2 &= \tfrac{3}{2} + t \\ s_3 &= t. \end{aligned}$$

Show that for any choice of the parameter t, the list (s_1, s_2, s_3) is a solution to the linear system

$$\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ x_1 + 3x_2 - x_3 &= 3. \end{aligned}$$

Solution. Substitute the list (s_1, s_2, s_3) into the left-hand-side of the first equation

$$\left(-\tfrac{3}{2} - 2t\right) + \left(\tfrac{3}{2} + t\right) + t = 0$$

and in the second equation

$$\left(-\tfrac{3}{2} - 2t\right) + 3(\tfrac{3}{2} + t) - t = -\tfrac{3}{2} + \tfrac{9}{2} = 3$$

Both equations are satisfied for any value of t. Because we can vary t arbitrarily, we get an infinite number of solutions parameterized by t. For example, compute the list (s_1, s_2, s_3) for $t = 3$ and confirm that the resulting list is a solution to the linear system. □

1.2 Matrices

We will use **matrices** to develop systematic methods to solve linear systems and to study the properties of the solution set of a linear system. Informally speaking, a **matrix** is an array or table consisting of *rows* and *columns*. For example,

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 7 & 11 & -5 \end{bmatrix}$$

is a matrix having $m = 3$ rows and $n = 4$ columns. In general, a matrix with m rows and n columns is a $m \times n$ matrix and the set of all such matrices will be denoted by $M_{m\times n}$. Hence, $\mathbf{A}$ above is a 3×4 matrix. The entry of $\mathbf{A}$ in the ith row and jth column will be denoted by a_{ij}. A matrix containing only one column is called a **column vector** and a matrix containing only one row is called a **row vector**. For example, here is a row vector

$$\mathbf{u} = \begin{bmatrix} 1 & -3 & 4 \end{bmatrix}$$

and here is a column vector

$$\mathbf{v} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

We can associate to a linear system three matrices: (1) the coefficient matrix, (2) the output column vector, and (3) the augmented matrix. For example, for the linear system

$$\begin{aligned} 5x_1 - 3x_2 + 8x_3 &= -1 \\ x_1 + 4x_2 - 6x_3 &= 0 \\ 2x_2 + 4x_3 &= 3 \end{aligned}$$

the coefficient matrix $\mathbf{A}$, the output vector $\mathbf{b}$, and the augmented matrix $[\mathbf{A}\ \mathbf{b}]$ are:

$$\mathbf{A} = \begin{bmatrix} 5 & -3 & 8 \\ 1 & 4 & -6 \\ 0 & 2 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix}, \quad [\mathbf{A}\ \mathbf{b}] = \begin{bmatrix} 5 & -3 & 8 & -1 \\ 1 & 4 & -6 & 0 \\ 0 & 2 & 4 & 3 \end{bmatrix}.$$

If a linear system has m equations and n unknowns then the coefficient matrix $\mathbf{A}$ must be a $m \times n$ matrix, that is, $\mathbf{A}$ has m rows and n columns. Using our previously defined notation, we can write this as $\mathbf{A} \in M_{m\times n}$.

If we are given an augmented matrix, we can write down the associated linear system in an obvious way. For example, the linear system associated to the augmented matrix

$$\begin{bmatrix} 1 & 4 & -2 & 8 & 12 \\ 0 & 1 & -7 & 2 & -4 \\ 0 & 0 & 5 & -1 & 7 \end{bmatrix}$$

is

$$\begin{aligned} x_1 + 4x_2 - 2x_3 + 8x_4 &= 12 \\ x_2 - 7x_3 + 2x_4 &= -4 \\ 5x_3 - x_4 &= 7. \end{aligned}$$

We can study matrices without interpreting them as coefficient matrices or augmented matrices associated to a linear system. **Matrix algebra** is a fascinating subject with numerous applications in every branch of engineering, medicine, statistics, mathematics, finance, biology, chemistry, etc.

1.3 Solving linear systems

In algebra, you learned to solve equations by first "simplifying" them using operations that do not alter the solution set. For example, to solve $2x = 8 - 2x$ we can add to both sides $2x$ and obtain $4x = 8$ and then multiply both sides by $\frac{1}{4}$ yielding $x = 2$. We can do similar operations on a linear system. There are three basic operations, called **elementary operations**, that can be performed:

1. Interchange two equations.
2. Multiply an equation by a nonzero constant.
3. Add a multiple of one equation to another.

These operations do not alter the solution set. The idea is to apply these operations iteratively to simplify the linear system to a point where one can easily write down the solution set. It is convenient to apply elementary operations on the augmented matrix $[\mathbf{A}\ \mathbf{b}]$ representing the linear system. In this case, we call the operations **elementary row operations**, and the process of simplifying the linear system using these operations is called **row reduction**. The goal with row reducing is to transform the original linear system into one having a **triangular structure** and then perform **back substitution** to solve the system. This is best explained via an example.

Example 1.5. Use back substitution on the augmented matrix

$$\begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

to solve the associated linear system.

Solution. Notice that the augmented matrix has a triangular structure. The third row corresponds to the equation $x_3 = 1$. The second row corresponds to the equation

$$x_2 - x_3 = 0$$

and therefore $x_2 = x_3 = 1$. The first row corresponds to the equation

$$x_1 - 2x_3 = -4$$

and therefore

$$x_1 = -4 + 2x_3 = -4 + 2 = -2.$$

Therefore, the solution is $(-2, 1, 1)$. □

Example 1.6. Solve the linear system using elementary row operations.

$$\begin{aligned} -3x_1 + 2x_2 + 4x_3 &= 12 \\ x_1 - 2x_3 &= -4 \\ 2x_1 - 3x_2 + 4x_3 &= -3 \end{aligned}$$

Solution. Our goal is to perform elementary row operations to obtain a triangular structure and then use back substitution to solve. The augmented matrix is

$$\begin{bmatrix} -3 & 2 & 4 & 12 \\ 1 & 0 & -2 & -4 \\ 2 & -3 & 4 & -3 \end{bmatrix}.$$

Interchange Row 1 (R_1) and Row 2 (R_2):

$$\begin{bmatrix} -3 & 2 & 4 & 12 \\ 1 & 0 & -2 & -4 \\ 2 & -3 & 4 & -3 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 0 & -2 & -4 \\ -3 & 2 & 4 & 12 \\ 2 & -3 & 4 & -3 \end{bmatrix}$$

As you will see, this first operation will simplify the next step. Add $3R_1$ to R_2:

$$\begin{bmatrix} 1 & 0 & -2 & -4 \\ -3 & 2 & 4 & 12 \\ 2 & -3 & 4 & -3 \end{bmatrix} \xrightarrow{3R_1+R_2} \begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 2 & -2 & 0 \\ 2 & -3 & 4 & -3 \end{bmatrix}$$

Add $-2R_1$ to R_3:

$$\begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 2 & -2 & 0 \\ 2 & -3 & 4 & -3 \end{bmatrix} \xrightarrow{-2R_1+R_3} \begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 2 & -2 & 0 \\ 0 & -3 & 8 & 5 \end{bmatrix}$$

Multiply R_2 by $\frac{1}{2}$:

$$\begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 2 & -2 & 0 \\ 0 & -3 & 8 & 5 \end{bmatrix} \xrightarrow{\frac{1}{2}R_2} \begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 1 & -1 & 0 \\ 0 & -3 & 8 & 5 \end{bmatrix}$$

Add $3R_2$ to R_3:

$$\begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 1 & -1 & 0 \\ 0 & -3 & 8 & 5 \end{bmatrix} \xrightarrow{3R_2+R_3} \begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 5 & 5 \end{bmatrix}$$

Multiply R_3 by $\frac{1}{5}$:

$$\begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 5 & 5 \end{bmatrix} \xrightarrow{\frac{1}{5}R_3} \begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

We can continue row reducing but the row reduced augmented matrix is in triangular form. So now use back substitution to solve. The linear system associated to the row reduced

augmented matrix is

$$\begin{aligned} x_1 - 2x_3 &= -4 \\ x_2 - x_3 &= 0 \\ x_3 &= 1 \end{aligned}$$

The last equation gives that $x_3 = 1$. From the second equation we obtain that $x_2 - x_3 = 0$, and thus $x_2 = 1$. The first equation then gives that $x_1 = -4+2(1) = -2$. Thus, the solution to the original system is $(-2, 1, 1)$. You should verify that $(-2, 1, 1)$ is a solution to the original system. □

The original augmented matrix of the previous example is

$$\mathbf{M} = \begin{bmatrix} -3 & 2 & 4 & 12 \\ 1 & 0 & -2 & -4 \\ 2 & -3 & 4 & -3 \end{bmatrix} \qquad \rightarrow \qquad \begin{aligned} -3x_1 + 2x_2 + 4x_3 &= 12 \\ x_1 - 2x_3 &= -4 \\ 2x_1 - 3x_2 + 4x_3 &= -3. \end{aligned}$$

After row reducing we obtained the row reduced matrix

$$\mathbf{N} = \begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \qquad \rightarrow \qquad \begin{aligned} x_1 - 2x_3 &= -4 \\ x_2 - x_3 &= 0 \\ x_3 &= 1. \end{aligned}$$

Although the two augmented matrices $\mathbf{M}$ and $\mathbf{N}$ are clearly distinct, it is a fact that they have the same solution set.

Example 1.7. Using elementary row operations, show that the linear system is inconsistent.

$$\begin{aligned} x_1 + 2x_3 &= 1 \\ x_2 + x_3 &= 0 \\ 2x_1 + 4x_3 &= 1 \end{aligned}$$

Solution. The augmented matrix is

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 2 & 0 & 4 & 1 \end{bmatrix}$$

Perform the operation $-2R_1 + R_3$:

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 2 & 0 & 4 & 1 \end{bmatrix} \xrightarrow{-2R_1+R_3} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

The last row of the simplified augmented matrix

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

corresponds to the equation

$$0x_1 + 0x_2 + 0x_3 = -1$$

Obviously, there are no numbers x_1, x_2, x_3 that satisfy this equation, and therefore, the linear system is inconsistent, i.e., it has no solution. In general, if we obtain a row in an **augmented matrix** of the form

$$\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & c \end{bmatrix}$$

where c is a **nonzero** number, then the linear system is inconsistent. We will call this type of row an **inconsistent row**. However, a row of the form

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

corresponds to the equation $x_2 = 0$ which is perfectly valid. □

1.4 Geometric interpretation of the solution set

The set of points (x_1, x_2) that satisfy the linear system

$$\begin{aligned} x_1 - 2x_2 &= -1 \\ -x_1 + 3x_2 &= 3 \end{aligned} \tag{1.2}$$

is the intersection of the two lines determined by the equations of the system. The solution for this system is $(3, 2)$. The two lines intersect at the point $(x_1, x_2) = (3, 2)$, see Figure 1.1.

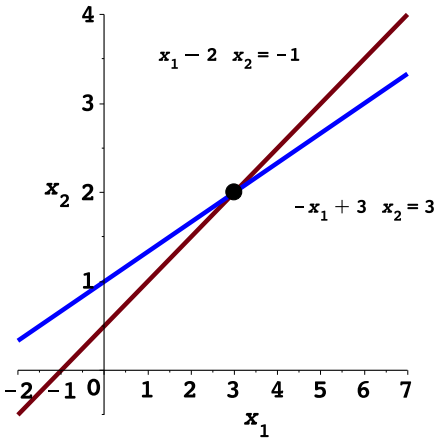


Figure 1.1: The intersection point of the two lines is the solution of the linear system (1.2)

Similarly, the solution of the linear system

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ -4x_1 + 5x_2 + 9x_3 &= -9 \end{aligned} \tag{1.3}$$

is the intersection of the three planes determined by the equations of the system. In this case, there is only one solution: $(29, 16, 3)$. In the case of a consistent system of two equations, the solution set is the line of intersection of the two planes determined by the equations of the system, see Figure 1.2.

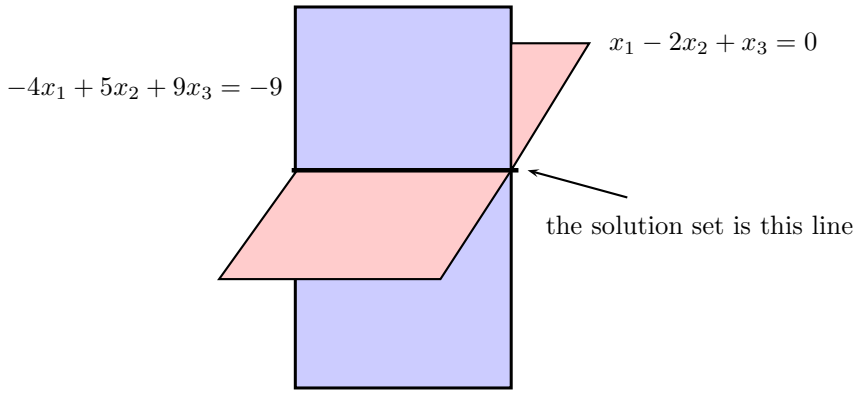


Figure 1.2: The intersection of the two planes is the solution set of the linear system (1.3)

After this lecture you should know the following:

- what a linear system is
- what it means for a linear system to be consistent and inconsistent
- what matrices are
- what are the matrices associated to a linear system
- what the elementary row operations are and how to apply them to simplify a linear system
- what it means for two matrices to be row equivalent
- how to use the method of back substitution to solve a linear system
- what an inconsistent row is
- how to identify using elementary row operations when a linear system is inconsistent
- the geometric interpretation of the solution set of a linear system